COUNTEREXAMPLE TO THE CLIQUE-CHROMATIC EQUALITY FOR GRAPH FRACTIONAL POWERS: $(C_3 \square K_2)^{3/5}$

ARGUS OPEN RESEARCH PROJECT

Abstract. For $1 < m < n$, the fractional power $G^{m/n}$ is the m-th distance power of the n-subdivision of G. We verify the counterexample of Hartke, Liu, and Petříčková (2012) to the conjecture that $\chi(G^{m/n}) = \omega(G^{m/n})$ for every connected graph with $\Delta(G) \geq 3$: for the triangular prism $H = C_3 \square K_2$ ($\Delta(H) = 3$) and $(m, n) = (3, 5)$, the 42-vertex, 162-edge graph $H^{3/5}$ has clique number $\omega(H^{3/5}) = 5$ and chromatic number $\chi(H^{3/5}) = 6$.

1. Main Proposition

Theorem 1.1. *Let $H = C_3 \square K_2$ be the 6-vertex, 9-edge triangular prism, and let $H^{3/5}$ be the third power of its 5-subdivision (42 vertices and 162 edges). Then:*

(1) *By Bron–Kerbosch clique enumeration (and the odd-m formula $\omega(G^{m/n}) = \frac{m-1}{2}\Delta(G) + 2$), $\omega(H^{3/5}) = 5$.*
(2) *Fixing a K_5 subgraph to colors $\{0, 1, 2, 3, 4\}$, exhaustive backtracking over all 5,185 valid partial 5-colorings finds zero proper 5-colorings of $H^{3/5}$, while an explicit proper 6-coloring exists; hence $\chi(H^{3/5}) = 6 > 5 = \omega(H^{3/5})$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3117 in the domain of graph coloring and fractional graph theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.
(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.
(c) *Classification:* Within the program taxonomy, this result is categorized as a *confirmed counterexample* deciding the scope: (C_3 box K_2)^{}{3/5} on 42 vertices and 162 edges has omega=5 and chi=6 > 5 (after Hartke-Liu-Petrickova 2012).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

References

[1] Primary problem list and source literature for Result 3117 (graph coloring and fractional graph theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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